VARYING "n" KEEPING OTHER PARAMETERS FIXED

Varying "n" at b=1 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	1	-14.5	45.6	23209420000	23297350000	FAILED RUN	369389400	370788900	FAILED RUN	0.1817224	0.2596383	FAILED RUN
2	1	-14.5	45.6	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN	FAILED RUN
3	1	-14.5	45.6	24234110000	FAILED RUN	FAILED RUN	385697800	FAILED RUN	FAILED RUN	0.1897453	FAILED RUN	FAILED RUN
4	1	-14.5	45.6	24241630000	FAILED RUN	24393250000	385817500	FAILED RUN	388230700	0.1898042	FAILED RUN	0.2718516
5	1	-14.5	45.6	FAILED RUN	FAILED RUN	24335660000	FAILED RUN	FAILED RUN	387314100	FAILED RUN	FAILED RUN	0.2712098
6	1	-14.5	45.6	24097480000	24259120000	24255030000	383523400	386095900	386030800	0.1886756	0.2707269	0.2703111
7	1	-14.5	45.6	FAILED RUN	24174660000	24170700000	FAILED RUN	384751600	384688600	FAILED RUN	0.2697843	0.2693713
8	1	-14.5	45.6	23930410000	FAILED RUN	24090880000	380864300	FAILED RUN	383418200	0.1873675	FAILED RUN	0.2684818
9	1	-14.5	45.6	FAILED RUN	24020500000	24016930000	FAILED RUN	382298200	382241300	FAILED RUN	0.268064	0.2676576
10	1	-14.5	45.6	FAILED RUN	23953350000	23950260000	FAILED RUN	381229300	381180200	FAILED RUN	0.2671997	0.2668

Varying "n" at b=2 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	2	-14.5	45.6	FAILED RUN	22583220000	FAILED RUN	FAILED RUN	359423200	FAILED RUN	FAILED RUN	0.2520242	FAILED RUN
2	2	-14.5	45.6	23726610000	FAILED RUN	23838250000	377620700	FAILED RUN	379397600	0.1857718	FAILED RUN	0.2656664
3	2	-14.5	45.6	24138160000	FAILED RUN	24271990000	384170800	FAILED RUN	386300800	0.1889941	FAILED RUN	0.2705002
4	2	-14.5	45.6	FAILED RUN	24394900000	24391820000	FAILED RUN	388256900	388207900	FAILED RUN	0.2722422	0.2718356
5	2	-14.5	45.6	24239320000	24392070000	24388640000	385780700	388211800	388211800	0.1897861	0.2722106	0.2718002
6	2	-14.5	45.6	24183230000	24339830000	24336340000	384888100	387380300	387324900	0.189347	0.2716276	0.2712174
7	2	-14.5	45.6	24111150000	24269720000	24266410000	383740800	386264500	386211800	0.1887826	0.2708452	0.2704379
8	2	-14.5	45.6	24034790000	24196040000	24192030000	382525600	385091900	385028100	0.1881848	0.270023	0.2696091
9	2	-14.5	45.6	23961040000	24122840000	24119400000	381351700	383926800	383872100	0.1876073	0.269206	0.2687996
10	2	-14.5	45.6	FAILED RUN	24054340000	24051140000	FAILED RUN	382836800	382785700	FAILED RUN	0.2684416	0.2680389

Varying "n" at b=3 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	3	-14.5	45.6	21717020000	21773510000	21149590000	345637100	346536200	336606200	0.1700373	0.242988	0.2357025
2	3	-14.5	45.6	23336880000	23439250000	23421780000	371417900	373047300	372769200	0.1827203	0.2615773	0.261025
3	3	-14.5	45.6	23961810000	24087470000	24082490000	381364000	383364000	383284700	0.1876133	0.2688114	0.2683882
4	3	-14.5	45.6	24186840000	24325680000	24322010000	384945500	387155200	387096800	0.1893753	0.2714697	0.2710576
5	3	-14.5	45.6	24241920000	24388350000	24384870000	385822100	388152700	388097300	0.1898065	0.2721691	0.2717582
6	3	-14.5	45.6	24222340000	24373530000	24370130000	385510500	387916700	387862700	0.1896532	0.2720037	0.2715939
7	3	-14.5	45.6	24169880000	24324470000	24321230000	384675700	387135900	387084300	0.1892425	0.2714562	0.2710489
8	3	-14.5	45.6	24105170000	24261990000	24258550000	383645700	386141600	386086900	0.1887358	0.270759	0.2703504
9	3	-14.5	45.6	24038100000	24195620000	24192080000	382578200	385085200	385028800	0.1882107	0.2700183	0.2696096
10	3	-14.5	45.6	23947580000	24107540000	24103900000	381137500	383683500	383625500	0.1875019	0.2690353	0.2686269

Varying "n" at b=4 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	4	-14.5	45.6	20937840000	20972210000	20001570000	333236100	333783100	318334800	0.1639366	0.2340456	0.2229083
2	4	-14.5	45.6	22803600000	22888730000	22849850000	362930600	364285500	363666700	0.178545	0.2554337	0.2546511
3	4	-14.5	45.6	23738180000	23852920000	23844410000	377804800	379631100	379495600	0.1858624	0.2661938	0.265735
4	4	-14.5	45.6	24080420000	24210760000	24206430000	383251800	385326200	385257300	0.188542	0.2701872	0.2697696
5	4	-14.5	45.6	24203490000	24343190000	24339630000	385210600	387433900	387377200	0.1895057	0.2716651	0.271254
6	4	-14.5	45.6	24224020000	24370260000	24366730000	385537300	387864700	387808600	0.1896664	0.2719672	0.271556
7	4	-14.5	45.6	24196270000	24346720000	24343330000	385095500	387490100	387436100	0.1894491	0.2717045	0.2712952
8	4	-14.5	45.6	24145990000	24299680000	24296290000	384295400	386741500	386687400	0.1890554	0.2711796	0.2707709
9	4	-14.5	45.6	24087000000	24242710000	24239370000	383356500	385834700	385781500	0.1885935	0.2705438	0.2701366
10	4	-14.5	45.6	24025210000	24182350000	24178930000	382373000	384874100	384819700	0.1881097	0.2698702	0.2694631

Varying "n" at b=5 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	5	-14.5	45.6	20215890000	20231320000	19111740000	321745800	321991500	304172800	0.1582839	0.2257775	0.2129916
2	5	-14.5	45.6	22481590000	22554110000	22373950000	357805600	358959800	356092400	0.1760237	0.2516993	0.2493473
3	5	-14.5	45.6	23497130000	23600970000	23587770000	373968400	375621200	375411000	0.183975	0.2633821	0.2628749
4	5	-14.5	45.6	23950200000	24072110000	24066850000	381179300	383119500	383035800	0.1875225	0.2686399	0.268214
5	5	-14.5	45.6	24139980000	24273500000	24269780000	384199700	386324700	386265500	0.1890083	0.2708874	0.2704755
6	5	-14.5	45.6	24202010000	24342630000	24339210000	385186900	387425100	387370500	0.189494	0.2716589	0.2712493
7	5	-14.5	45.6	24200000000	24346220000	24342620000	385154900	387482100	387424900	0.1894783	0.2716989	0.2712873
8	5	-14.5	45.6	24165430000	24316330000	24312860000	384604800	387006400	386951100	0.1892076	0.2713653	0.2709556
9	5	-14.5	45.6	24117420000	24270110000	24266760000	383840700	386270800	386217500	0.1888317	0.2708495	0.2704419
10	5	-14.5	45.6	24062030000	24219040000	24213610000	382959000	385458000	385371600	0.188398	0.2702796	0.2698496

Varying "n" at b=6 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	6	-14.5	45.6	19571290000	19571080000	18410200000	311486800	311483400	293007500	0.153237	0.2184093	0.2051732
2	6	-14.5	45.6	22077080000	22137060000	21821450000	351367600	352322200	347299200	0.1728565	0.2470451	0.24319
3	6	-14.5	45.6	23255880000	23350430000	23330250000	370128900	371633600	371312500	0.1820862	0.2605861	0.2600049
4	6	-14.5	45.6	23810130000	23925250000	23918110000	378949900	380782100	380668500	0.1864257	0.2670009	0.2665563
5	6	-14.5	45.6	24063270000	24191100000	24186730000	382978800	385013300	384943700	0.1884077	0.2699678	0.26955
6	6	-14.5	45.6	24164890000	24301330000	24298030000	384596200	386767700	386715100	0.1892034	0.271198	0.2707903
7	6	-14.5	45.6	24135270000	24276550000	24273390000	384124800	386373200	386323000	0.1889715	0.2709214	0.2705158
8	6	-14.5	45.6	24172080000	24318130000	24315570000	384710600	387035100	386994300	0.1892597	0.2713855	0.2709859
9	6	-14.5	45.6	24134590000	24285220000	24280970000	384113900	386511400	386443700	0.1889662	0.2710182	0.2706003
10	6	-14.5	45.6	24086530000	24238890000	24235460000	383349000	385773800	385719300	0.1885898	0.2705011	0.2700931

Varying "n" at b=7 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	7	-14.5	45.6	19005090000	18991950000	17847810000	302475400	302266300	284056800	0.1488038	0.2119464	0.1989057

2	7	-14.5	45.6	21703420000	21752620000	21338740000	345420600	346203700	339616500	0.1699308	0.2427549	0.2378104
3	7	-14.5	45.6	23023920000	23110120000	23072730000	366437100	367808900	367214000	0.18027	0.2579043	0.257135
4	7	-14.5	45.6	23669550000	23778330000	23769100000	376712600	378443900	378297000	0.185325	0.2653614	0.2648957
5	7	-14.5	45.6	23980720000	24103620000	24098550000	381665100	383621100	383540300	0.1877614	0.2689916	0.2685672
6	7	-14.5	45.6	24120250000	24252620000	24248730000	383885800	385992500	385930500	0.1888539	0.2706544	0.2702409
7	7	-14.5	45.6	24168750000	24307730000	24304080000	384657600	386869500	386811400	0.1892336	0.2712694	0.2708578
8	7	-14.5	45.6	24168660000	24312430000	24308770000	384656200	386944300	386886100	0.1892329	0.2713218	0.2709101
9	7	-14.5	45.6	24142200000	24289730000	24286270000	384235100	386583000	386527900	0.1890258	0.2710685	0.2706593
10	7	-14.5	45.6	24102030000	24252570000	24249890000	383595700	385991600	385949000	0.1887112	0.2706538	0.2702539

Varying "n" at b=8 (other parameters fixed)

n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	8	-14.5	45.6	18509710000	18486070000	17381990000	294591200	294215000	276643000	0.1449251	0.2063009	0.1937143
2	8	-14.5	45.6	21362140000	21402090000	20919200000	339989000	340624900	332939500	0.1672588	0.2388431	0.2331349
3	8	-14.5	45.6	22804950000	22883750000	22812840000	362952000	364206200	363077600	0.1785555	0.255378	0.2542386
4	8	-14.5	45.6	23532640000	23635290000	23624300000	374533600	376167300	375992300	0.1842531	0.2637651	0.2632819
5	8	-14.5	45.6	23897100000	24015230000	24009460000	380334200	382214300	382122300	0.1871067	0.2680052	0.2675743
6	8	-14.5	45.6	24071350000	24200180000	24195970000	383107400	385157800	385090800	0.188471	0.2700691	0.269653
7	8	-14.5	45.6	24143480000	24279250000	24275440000	384255400	386416200	386355600	0.1890357	0.2709515	0.2705386
8	8	-14.5	45.6	24158950000	24300090000	24296820000	384501700	386747900	386695900	0.1891569	0.2711841	0.2707769
9	8	-14.5	45.6	24143620000	24288690000	24285340000	384257700	386566500	386513200	0.1890369	0.2710569	0.270649
10	8	-14.5	45.6	24111040000	24258970000	24255670000	383739100	386093400	386041000	0.1887818	0.2707252	0.2703183

Varying "n" at b=9 (other parameters fixed)

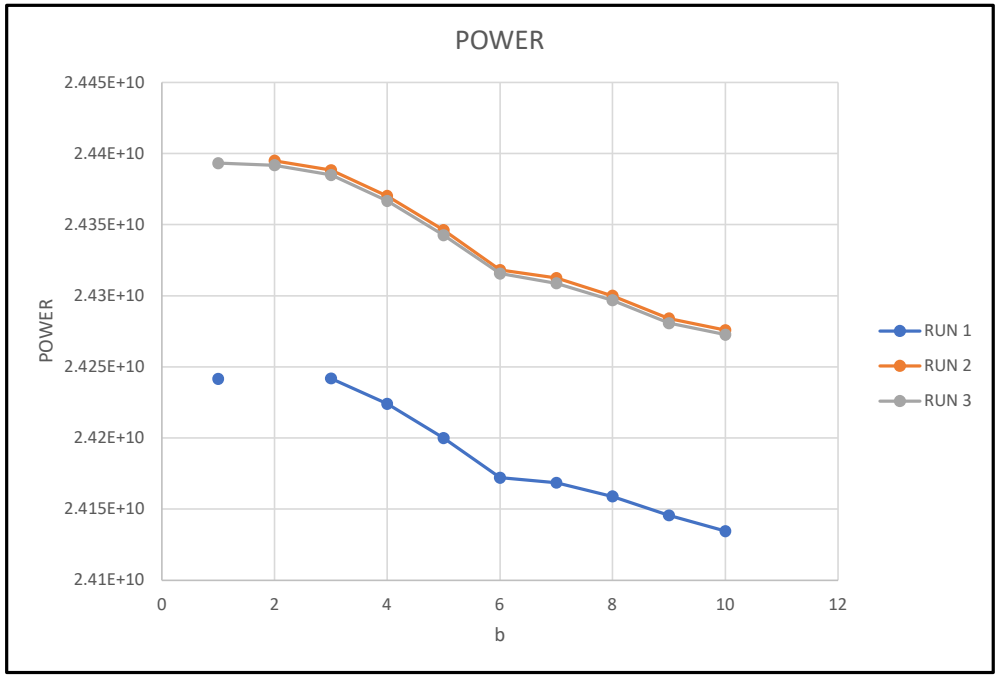
n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	9	-14.5	45.6	18075090000	18042770000	16990540000	287674000	287159500	270412800	0.1415222	0.2013537	0.1893518
2	9	-14.5	45.6	21051060000	21082840000	20558240000	335038100	335543800	327194600	0.1648231	0.2352803	0.2291122
3	9	-14.5	45.6	22599970000	22672140000	22566760000	359689700	360838300	359161100	0.1769506	0.2530166	0.2514962
4	9	-14.5	45.6	23400840000	23498310000	23484000000	372435900	373987200	373759500	0.1832211	0.2622364	0.2617184
5	9	-14.5	45.6	23814230000	23928090000	23921480000	379015200	380827300	380722200	0.1864578	0.2670327	0.2665939
6	9	-14.5	45.6	24020910000	24145940000	24141550000	382304600	384294600	384224800	0.1880761	0.2694638	0.2690465
7	9	-14.5	45.6	24114660000	24247850000	24244000000	383796800	385916500	385855200	0.1888101	0.2706011	0.2701882
8	9	-14.5	45.6	24145590000	24284150000	24280790000	384289000	386494200	386440800	0.1890523	0.2710062	0.2705982
9	9	-14.5	45.6	24140870000	24283550000	24280130000	384213900	386484800	386430300	0.1890153	0.2709996	0.2705909
10	9	-14.5	45.6	24115510000	24261640000	24258360000	383810300	386136000	386083900	0.1888168	0.270755	0.2703483

Varying "n" at b=10 (other parameters fixed)

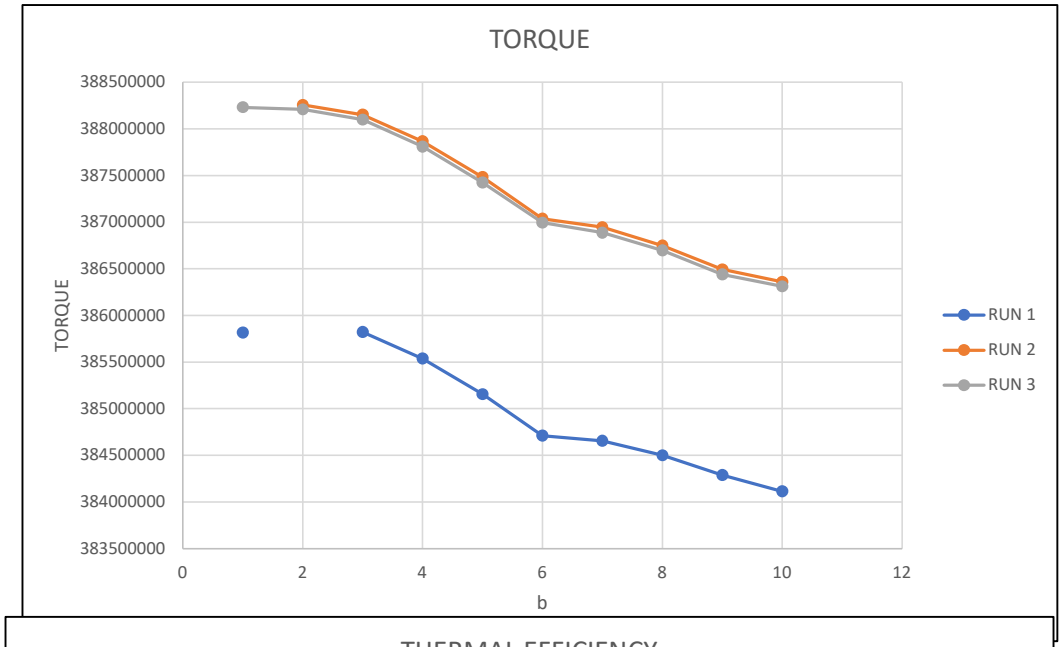
n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
1	10	-14.5	45.6	17691720000	17652080000	16655500000	281572500	280941500	265080500	0.1385206	0.1969936	0.1856179
2	10	-14.5	45.6	20766960000	20791700000	20239890000	330516400	330910100	322127800	0.1625987	0.2320312	0.2255642
3	10	-14.5	45.6	22408420000	22474610000	22333390000	356641100	357694500	355446900	0.1754508	0.2508121	0.2488953
4	10	-14.5	45.6	23275260000	23367790000	23352220000	370437200	371909900	371662200	0.1822379	0.2607798	0.2602498

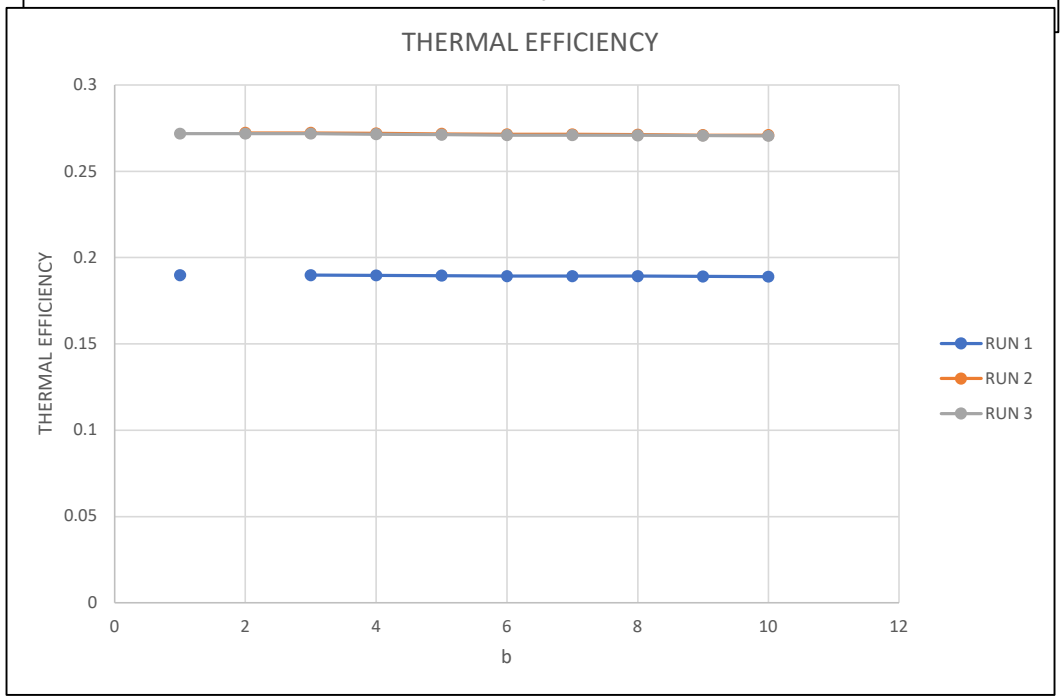
5	10	-14.5	45.6	23733010000	23843080000	23835990000	377722500	379474300	379361600	0.1858219	0.2660839	0.2656412
6	10	-14.5	45.6	23970060000	24092120000	24087400000	381495300	383438000	383362900	0.1876779	0.2688633	0.268443
7	10	-14.5	45.6	24084590000	24214640000	24211030000	383318200	385388000	385330600	0.1885747	0.2702306	0.2698208
8	10	-14.5	45.6	24129670000	24265900000	24262460000	384035600	386203800	386149000	0.1889276	0.2708026	0.2703939
9	10	-14.5	45.6	24134630000	24275760000	24272730000	384114500	386360800	386312500	0.1889664	0.2709127	0.2705084
10	10	-14.5	45.6	24116370000	24262370000	24257920000	383824000	386147600	386076800	0.1888235	0.2707632	0.2703433

Optimal "n" and "b"												
n	b	SOC	Burn Duration	POWER (erg/sec)			TORQUE (dyne-cm)			THERMAL EFFICIENCY		
				RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3	RUN 1	RUN 2	RUN 3
4	1	-14.5	45.6	24241630000		24393250000	385817500		388230700	0.1898042		0.2718516
4	2	-14.5	45.6		24394900000	24391820000		388256900	388207900		0.2722422	0.2718356
5	3	-14.5	45.6	24241920000	24388350000	24384870000	385822100	388152700	388097300	0.1898065	0.2721691	0.2717582
6	4	-14.5	45.6	24224020000	24370260000	24366730000	385537300	387864700	387808600	0.1896664	0.2719672	0.271556
7	5	-14.5	45.6	24200000000	24346220000	24342620000	385154900	387482100	387424900	0.1894783	0.2716989	0.2712873
8	6	-14.5	45.6	24172080000	24318130000	24315570000	384710600	387035100	386994300	0.1892597	0.2713855	0.2709859
8	7	-14.5	45.6	24168660000	24312430000	24308770000	384656200	386944300	386886100	0.1892329	0.2713218	0.2709101
8	8	-14.5	45.6	24158950000	24300090000	24296820000	384501700	386747900	386695900	0.1891569	0.2711841	0.2707769
8	9	-14.5	45.6	24145590000	24284150000	24280790000	384289000	386494200	386440800	0.1890523	0.2710062	0.2705982
9	10	-14.5	45.6	24134630000	24275760000	24272730000	384114500	386360800	386312500	0.1889664	0.2709127	0.2705084



Optimal values:
n = 4
b = 2


TORQUE
388500000
388000000
387500000
387000000
386500000
386000000
385500000
385000000
384500000
384000000
383500000
TORQUE
0
2
4
6
8
10
12
b
RUN 1
RUN 2
RUN 3


THERMAL EFFICIENCY
0.3
0.25
0.2
0.15
0.1
0.05
0
THERMAL EFFICIENCY
0
2
4
6
8
10
12
b
RUN 1
RUN 2
RUN 3